Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/03/2020
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Project Title:	Project Atlas		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77003	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	05/01/2019	**End Date:**	04/30/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0007		
Performance Goal No.:			
Performance Goal Text:			
	Industry Project Z3VR is proposing the development of a virtual reality (VR) game that is capable of enhancing pre-existing physical maintenance routines, managing stress, and serving as a behavioral health countermeasure. We hypothesize that we can enhance the benefits of meditative respiration and exercise through visual biofeedback loops in integrated VR environments. We plan to design an adaptive application with two complementary modules (exercise and meditation) built to enhance conscious control of the autonomic nervous system. The game will use real time biometric data (heart rate and respiration rate) to influence key game play properties (interactable objects, "non-playable character" (NPC) behavior, difficulty, intensity, lighting). In the exercise module, users will perform physically challenging tasks in order to increase their heart rate into a desired range. A user's ability to modulate their own heart rate will influence their performance in		

Task Description:	this module. Exercise is an effective tool for managing stress, anxiety, and depression (Paolucci et al., 2018, Gujral et al., 2017, and Craft and Perna, 2004). VR enables exercise variety, which has been shown to be an important factor in the adherence of physical maintenance routines. VR has also been shown to enhance the overall performance and impact of physical exercise (Gujral et al., 2017), and amplifies it s mood-boosting benefits. In the meditation module the game will use respiration rate data gathered from a microphone built into a VR head-mounted-display (HMD) in order to facilitate the optimal meditation session. In this module the properties of the user's environment (color palette, saturation, etc.). Specifically, the game will provide visual and auditory feedback toward successfully lowering heart rate. The game will also track the sessions of each individual player over time. The player's history and personalization preferences (e.g., target heart rate, session length) will be used to determine the difficulty and scoring of each session. Paolucci T, Attanasi C, Cecchini W, Marazzi A, Capobianco SV, Santilli V. Chronic low back pain and postural rehabilitation exercise: a literature review. J Pain Res. 2018;12:95-107. Published 2018 Dec 20. https:// Gujral S, Aizenstein H, Reynolds CF 3rd, Butters MA, Erickson KI. Exercise effects on depression: Possible neural mechanisms. Gen Hosp Psychiatry. 2017;49:2-10. https:// Craft LL, Perna FM. The benefits of exercise for the clinically depressed. Prim Care Companion J Clin Psychiatry. 2004;6(3):104-111. https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research tool enables future studies on the efficacy of using VR-based exercise and meditation modules to address behavioral health risks associated with the isolation, confinement, and stimulus deprivation related to exploration class space missions. Our tool is a first step toward a data-driven, real-time behavioral health countermeasure, exercise prescription, and form-monitoring platform. Collecting, graphing, and distributing exercise data and providing real-time form correction could revolutionize existing Astronaut Strength, Conditioning and Rehabilitation (ASCR) workflows, as form correction in particular currently presents significant challenges due to the slow delivery of workout footage to analyze during which crewmembers are more susceptible to injury. From a content standpoint, this validates the feasibility of high-fidelity VR content that responds in real time to changes in the subject's physiological signals. This project uses features (custom environments, task gamification, and a relaxation module) which call for further research on their efficacy in addressing deep spaceflight-related cognitive health. Further work will be required to make adaptive experiences that can sustain the physical and mental health of the crew during long periods of limited contact with the outside, and this project's approach of reacting to passive input from the user's body will likely be a key element of the design of any eventual solution to this problem, as described elsewhere in our report. Additionally, this project can serve as an important step in advancing space-ready VR hardware specifications. Our Product Requirements and Specifications deliverable outlines the technology requirements for a VR-based behavioral and physical fitness tool to operate in exploration class missions. Lessons learned from integrating NASA exercise equipment will also facilitate inclusion of other existing or future devices for deep spaceflight.
Task Progress:	Background Long-term spaceflight will place enormous physical and psychological stress on crewmembers through prolonged confinement, exposure to microgravity, high mission uncertainty, noise, lack of privacy, high-latency communication with Earth, and the boredom of repetitive environments and tasks. Whereas current mid-flight countermeasures are designed to maintain crew mission capabilities and allow for successful rehabilitation once on Earth, long-duration countermeasures will require greater autonomy, variety, and recreational elements in order to remain effective throughout the mission. Virtual Reality (VR) technology has shown promise as a component of future countermeasures. Studies have shown VR is used to enhance subject performance and impact for physical exercise, and to amplify its mood-boosting benefits. Additionally, VR enables exercise variety, which is an important factor in adherence to physical maintenance routines. Original aims Our initial aim was to investigate the feasibility and impact of a VR fitness tracking application built to support current countermeasures against confinement, isolation, lack of privacy and novel stimuli, and muscle and bone density loss. For our phase 1 grant, we proposed developing a research-ready exercise and meditation VR app that featured real-time feedback from networked biosensors. Outcomes As a research tool, we have built an adaptive application with exercise and breathing modules that uses real-time biometric data (heart rate and respiration rate) to influence key gameplay properties (e.g., difficulty, intensity, lighting, and environmental objects). The exercise module optionally interfaces with NASA exercise hardware to provide telemetry data, and then records detailed subject performance locally and remotely for analysis. The breathing module provides an environment that responds to changes in respiration rate and heart rate as proxies for relaxation measures and effects, respectively. Additionally, our Product Requirements and Specifications deliverable outlines the technology requirements for a VR-based behavioral and physical fitness tool to operate in exploration class missions. Next steps We plan to use our platform to study the impact of a virtual reality component on behavioral and physical health countermeasures. We are currently engaging research and private partners about the design of a possible isolation study about VR- and non-VR-enabled countermeasures' effects on individual crewmembers' biological signals, cognitive performance, and team cohesion.
Bibliography Type:	Description: (Last Updated:)